CMSC 420: Blockchains

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January 21, 2026

1 Problems **2**

1 Problems

1. The following is a really basic (and pretty terrible) simple version of SHA-256. It only works on keys M which are binary strings whose length is a multiple of 4.

   (a) Break M up into N sub-keys of length 4 bits: $M^{(1)}$, $M^{(2)}$, ... , $M^{(N)}$.

   (b) Initialize $H = \texttt{0101}$ and $K = \texttt{0111}$.

   (c) For each sub-key $M^{(i)}$ do the following:

      i. Update H by XORing it with $M^{(i)}$.

      ii. Update H by XORing it with K.

      iii. Left-rotate K by 1 bit.

   (d) The output is the resulting H, so $h(M) = H$.

   (a) Calculate $h(\texttt{11110100})$. Show the value of H after each step.

   (b) Calculate $h(\texttt{00000000})$. Show the value of H after each step.

   (c) Find the smallest 8-bit nonce which can be prepended to `10101010` so that the output starts with at exactly two `0`s. Do this any way you wish but justify your approach.

2. In this problem we construct a small blockchain using the hash function from the previous question. We won't worry about including the date.

   (a) The first block contains the message `LOVE`. Convert this to ASCII binary. This is our first block.

   (b) The second block contains the message `YOU`. Convert this to ASCII binary, hash the first block, append that hash to the message. This is our second block.

   (c) The third block contains the message `CS`. Convert this to ASCII binary, hash the second block, append that hash to the message. This is our third block.

3. Suppose we have the following set of transactions. Note that they don't all have the same number of bits, which is not a problem:

$$111111, 010101010, 110100000, 001011010, 111011, 010$$

Use the very simple hash function h which breaks a binary string into chunks of 3 bits and XORs them to create the Merkle tree for this set of transactions.

4. Suppose have the following Merkle tree which was created using the very simple hash function h which breaks a binary string into chunks of 3 bits and XORs them:

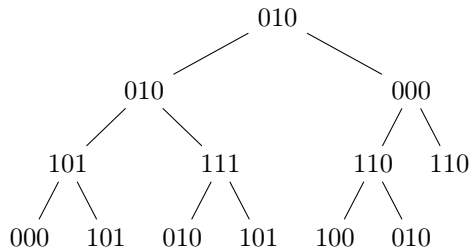

Figure 1: Merkle Tree

Provide a Merkle proof that the transaction 111011 is a valid transaction.

5. One thing we glossed over is if two transactions have the same hash (very unlikely unless they're the same transaction) then the concept of a Merkle proof presents an issue. Explain.

6. Suppose a Merkle tree has n keys. For each of the following n, how many nodes will be in the Merkle tree?

n	Number of Nodes
5	
6	
2	
5	
17	
11	

7. Suppose a Merkle tree has n keys. If $f(n)$ is the number of nodes in the Merkle tree, prove that:

$$f(n) = \begin{cases} 1 & \text{if } n = 1 \\ 2\lceil n/2 \rceil + f(\lceil n/2 \rceil) & \text{otherwise} \end{cases}$$